CORE JAVA – 5 MIN FINAL REVISION

🔴 🔥 BIG PROGRAM PATTERN (Q1,3,5,7,9,11,13,15)

1. Create Base Class
2. Create Child Classes (extends)
3. Choose Collection:
   - HashSet → unique ids
   - ArrayList → normal list
   - LinkedList → same as list
   - HashMap / TreeMap → key-value
   - TreeSet → sorted

4. Menu (while + switch)
5. Operations:
   - Add → add()/put()
   - Remove → remove()/removeIf()
   - Update → loop + modify
   - Display → for-each loop

🔴 COLLECTION QUICK MAP

HashSet → no duplicates
ArrayList → normal list
LinkedList → same list
HashMap → key-value
TreeMap → sorted keys
TreeSet → sorted elements
LinkedHashMap → insertion order

🔴 STRING QUESTIONS PATTERN (Q2,4,6,8,10,12,14,16)

1. Take input in List
2. Create list2, list3
3. Loop each string
4. Apply condition

5. Add to list2/list3
6. Print
7. Write to file (FileWriter)

🔴 IMPORTANT CONDITIONS

Vowel:
"aeiouAEIOU".indexOf(c) != -1

Only digits:
s.matches("\\d+")

Uppercase:
s.equals(s.toUpperCase())

Contains:
s.toLowerCase().contains("java")

Palindrome:
s.equals(new StringBuilder(s).reverse().toString())

Same start & end:
s.charAt(0) == s.charAt(s.length()-1)

Contains space:
s.contains(" ")

No vowel:
if("aeiou".indexOf(c) == -1)

Repeated char:
nested loop compare

🔴 FILE WRITING

```
FileWriter fw = new FileWriter("file.txt");
for(String s : list) {
```

```
    fw.write(s + "\n");
}
fw.close();
```

🔴 Q17 – LONGEST SUBSTRING

start from each index
build string
if duplicate → break
track max length

🔴 Q18 – LONGEST PALINDROME

generate substring
reverse
compare
store longest

🔴 Q19 – PRODUCT ARRAY

for each element:
multiply all except itself
store in result[]

🔴 Q20 – MATRIX ROTATION

STEP 1: Transpose
STEP 2: Reverse each row

🔴 IMPORTANT JAVA SHORTCUTS

```
// remove using condition
list.removeIf(x -> x.id == id);

// loop
```

```
for(Object o : list)

// get from map
map.get(key)

// add to map
map.put(key, value)
```

🔴 **EXAM STRATEGY (VERY IMPORTANT)**

1. Start with STRING question (easy marks)
2. Then BIG program
3. Then ONE logic (Q17/Q19/Q20)

Don’t get stuck
Write partial code if needed